RESTRICTION TO A SMALLER BIG SITE PRESERVES SCHEMES AND ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement contains an align block displaying the two morphisms of topoi; it is part of the counted statement span and is preserved verbatim.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Description of the change-of-big-site morphisms of topoi. Topologies, Section topologies-section-change-alpha and Lemma topologies-lemma-change-alpha, used at proof lines 2776-2789 and 2805-2807. Supplies f^{-1} and the identification f^{-1}h_X = h'_X.
- Presentation of an algebraic space. Lemma lemma-space-presentation (1387-1407, produced as L44183), used at proof lines 2810-2813. Writes F = h_U/h_R.
- The presentation theorem for algebraic spaces. Theorem theorem-presentation, used at proof lines 2814-2815. Concludes that f^{-1}F is an algebraic space.

Source reference. Original lemma 0262: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2835–2835. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 02WW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3819–3819. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 022K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2709–2709. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 022I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2707–2707. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Suppose given big sites Sch_{fppf} and Sch'_{fppf}. Assume that Sch_{fppf} is contained in Sch'_{fppf}, see Topologies, Section 022I. Let S be an object of Sch_{fppf}. Let*

$$g : Sh((Sch/S)_{fppf}) \longrightarrow Sh((Sch'/S)_{fppf}),$$
$$f : Sh((Sch'/S)_{fppf}) \longrightarrow Sh((Sch/S)_{fppf})$$

be the morphisms of topoi of Topologies, Lemma 022K. Let F be a sheaf of sets on $(Sch/S)_{fppf}$. Then

1. *if F is representable by a scheme $X \in \mathrm{Ob}((Sch/S)_{fppf})$ over S, then $f^{-1}F$ is representable too, in fact it is representable by the same scheme X, now viewed as an object of $(Sch'/S)_{fppf}$, and*
2. *if F is an algebraic space over S, then $f^{-1}F$ is an algebraic space over S also.*

Proof. Let $X \in \mathrm{Ob}((Sch/S)_{fppf})$. Let us write h_X for the representable sheaf on $(Sch/S)_{fppf}$ associated to X, and h'_X for the representable sheaf on $(Sch'/S)_{fppf}$ associated to X. By the description of f^{-1} in Topologies, Section 022I we see that $f^{-1}h_X = h'_X$. This proves (1).

Next, suppose that F is an algebraic space over S. By Lemma 0262 this means that $F = h_U/h_R$ for some étale equivalence relation $R \to U \times_S U$ in $(Sch/S)_{fppf}$. Since f^{-1} is an exact functor we conclude that $f^{-1}F = h'_U/h'_R$. Hence $f^{-1}F$ is an algebraic space over S by Theorem 02WW. □